

6(a)	- p.d. across capacitor proportional to charge on capacitor - p.d. across capacitor = p.d. across resistor - current in resistor proportional to p.d. across resistor - current in resistor = rate of decrease of charge on capacitor <i>Any two points, 1 mark each</i>	B2
	charge proportional to current so rate of decrease of current decreases as current decreases (therefore exponential shape)	B1
6(b)(i)	$R = V / I$ $= 12 / (0.13 \times 10^{-3})$	C1
	$= 9.2 \times 10^4 \Omega$	A1
6(b)(ii)	correct read-off of at least one pair of values for I and t	C1
	attempted read-off of t when $I = 0.048$ mA or substitution of a correct pair of values of I and t into $I = 0.13 \exp(-t / \tau)$	C1
	$\tau = 4.3$ s	A1
6(c)	$\tau = RC$	C1
	$C = \tau / R = 4.3 / (9.2 \times 10^4)$ $= 4.7 \times 10^{-5}$ F	A1